LDA vs PCA Comparison MCQs

1. Which statement correctly distinguishes PCA from LDA?

- o A) PCA is supervised, while LDA is unsupervised
- o B) PCA is unsupervised, while LDA is supervised
- o C) Both are supervised techniques
- o D) Both are unsupervised techniques

Answer: B) PCA is unsupervised, while LDA is supervised

2. What information does PCA primarily use to determine the projection directions?

- o A) Class labels
- o B) Class means
- o C) Variance in the data
- o D) Class probabilities

Answer: C) Variance in the data

3. What information does LDA use that PCA does not?

- o A) Feature values
- o B) Class labels
- o C) Number of samples
- o D) Number of features

Answer: B) Class labels

4. The primary objective of PCA is to:

- o A) Maximize class separation
- o B) Minimize within-class variance
- o C) Maximize variance in the projected data
- o D) Maximize classification accuracy

Answer: C) Maximize variance in the projected data

5. The primary objective of LDA is to:

- o A) Maximize total variance
- o B) Maximize between-class separation while minimizing within-class variation
- o C) Minimize the number of samples
- o D) Remove all correlated features

Answer: B) Maximize between-class separation while minimizing within-class variation

6. Which technique is generally more suitable when the main goal is classification-oriented dimensionality reduction?

- o A) PCA
- o B) LDA
- o C) K-Means
- o D) Apriori

Answer: B) LDA

7. Which technique can be applied when class labels are unavailable?

- o A) LDA only
- o B) PCA only
- o C) Both PCA and LDA
- o D) Neither PCA nor LDA

Answer: B) PCA only

8. Consider a dataset where the direction of maximum variance does not correspond to the direction that best separates the classes. Which method is more likely to find the class-separating direction?

- o A) PCA
- o B) LDA
- o C) K-Means
- o D) Linear Regression

Answer: B) LDA

9. Which matrices are central to LDA but not required in the same way for standard PCA?

- o A) Within-class and between-class scatter matrices
- o B) Adjacency and distance matrices
- o C) Precision and recall matrices
- o D) Confusion and covariance matrices

Answer: A) Within-class and between-class scatter matrices

10. PCA generally finds principal components by performing:

- o A) Eigenvalue decomposition of the covariance matrix
- o B) Classification using class means
- o C) Clustering based on labels
- o D) Regression on target variables

Answer: A) Eigenvalue decomposition of the covariance matrix

11. LDA commonly obtains discriminant directions using:

- o A) $S_W^{-1}S_B$
- o B) $S_B^{-1}S_W$ only
- o C) The identity matrix
- o D) The confusion matrix

Answer: A) $S_W^{-1}S_B$

12. For a dataset with C classes, the maximum number of meaningful LDA components is:

- o A) C
- o B) C + 1
- o C) C - 1
- o D) Equal to the number of features

Answer: C) C - 1

13. The maximum number of principal components produced by PCA is generally related to:

- o A) Number of classes
- o B) Number of features
- o C) Number of labels
- o D) Number of class means

Answer: B) Number of features

14. Which method attempts to preserve the directions containing the greatest overall variance?

- o A) LDA
- o B) PCA
- o C) Both always
- o D) Neither

Answer: B) PCA

15. Which method explicitly attempts to reduce within-class scatter?

- o A) PCA
- o B) LDA
- o C) Both
- o D) Neither

Answer: B) LDA

16. Suppose two classes have very different means but their largest variance occurs in a direction where the classes overlap heavily. Which method is more likely to prioritize class separation?

- o A) PCA
- o B) LDA
- o C) Both equally
- o D) Neither

Answer: B) LDA

17. Which statement about PCA and LDA is correct?

- o A) PCA always produces better classification accuracy than LDA
- o B) LDA always produces better classification accuracy than PCA
- o C) Their performance depends on the characteristics and assumptions of the dataset
- o D) PCA and LDA are mathematically identical

Answer: C) Their performance depends on the characteristics and assumptions of the dataset

18. Which technique is more affected by the absence of class labels?

- o A) PCA
- o B) LDA
- o C) Both equally
- o D) Neither

Answer: B) LDA

19. A researcher wants to visualize a labeled dataset in 2D while maximizing separation between different classes. Which method would generally be the more appropriate first choice?

- o A) PCA
- o B) LDA
- o C) K-Means
- o D) Apriori

Answer: B) LDA

20. A researcher wants to reduce the dimensionality of an unlabeled dataset while retaining as much overall variance as possible. Which method should be preferred?

- o A) LDA
- o B) PCA
- o C) Logistic Regression
- o D) QDA

Answer: B) PCA

21. Which statement is NOT correct?

- o A) PCA does not require class labels
- o B) LDA requires class labels
- o C) PCA directly maximizes class separability
- o D) LDA considers class information

Answer: C) PCA directly maximizes class separability

22. If the dataset contains 3 classes, what is the maximum number of LDA projection dimensions?

- o A) 1
- o B) 2
- o C) 3
- o D) Equal to the number of original features

Answer: B) 2

23. Which method is generally more appropriate as a preprocessing technique when the goal is noise reduction without using class labels?

o A) LDA
o B) PCA
o C) QDA
o D) Logistic Regression

Answer: B) PCA

24. Which comparison is correct?

o A) PCA -> maximize variance; LDA -> maximize class separability
o B) PCA -> maximize class separability; LDA -> maximize variance
o C) PCA -> supervised; LDA -> unsupervised
o D) PCA -> classification only; LDA -> clustering only

Answer: A) PCA -> maximize variance; LDA -> maximize class separability

25. If the direction of maximum variance in a dataset is unrelated to class discrimination, what is the main potential disadvantage of using PCA for classification?

o A) PCA may preserve high-variance features that do not help distinguish classes
o B) PCA cannot perform dimensionality reduction
o C) PCA requires class labels
o D) PCA always removes the class-discriminative information

Answer: A) PCA may preserve high-variance features that do not help distinguish classes

26. Which technique is more directly designed to find a projection where samples belonging to the same class are close and different classes are far apart?

o A) PCA
o B) LDA
o C) K-Means
o D) Standardization

Answer: B) LDA

27. Which statement best summarizes the difference between PCA and LDA?

o A) PCA focuses on data variance, whereas LDA focuses on class discrimination
o B) PCA focuses on class discrimination, whereas LDA focuses on data variance
o C) Both focus only on reducing computational complexity
o D) Both require class labels and optimize the same objective

Answer: A) PCA focuses on data variance, whereas LDA focuses on class discrimination

28. In a classification problem, which combination can be useful when the original feature space has a very high number of dimensions?

- o A) PCA followed by LDA
- o B) LDA followed by PCA only
- o C) PCA followed by K-Means only
- o D) PCA followed by data deletion

Answer: A) PCA followed by LDA

29. Why might PCA be applied before LDA in a high-dimensional dataset?

- o A) To increase the number of classes
- o B) To reduce dimensionality and help address singularity of the within-class scatter matrix
- o C) To remove class labels
- o D) To convert the problem into clustering

Answer: B) To reduce dimensionality and help address singularity of the within-class scatter matrix

30. Which pair is correctly matched?

- o A) PCA — Unsupervised — Variance Maximization
- o B) PCA — Supervised — Class Separation
- o C) LDA — Unsupervised — Variance Maximization
- o D) LDA — Unsupervised — Clustering

Answer: A) PCA — Unsupervised — Variance Maximization